

SPECIFICATION OF SOUND ABSORPTION PERFORMANCE OF INTERIOR MATERIAL FINISHES

I. GENERAL

Provide sound absorptive materials for room to control sound reverberant field.

II. Reverberation Time Criteria

The Reverberation Time (RT) of the following spaces shall be as follows:

Space	RT at mid frequencies (500Hz, 1kHz and 2kHz)
Office type premises: Security Control Room & Management Offices / Network Operation Centre /Office	< 0.6 second
Conference Rooms	

The above RT criteria is in accordance to the requirements of HKBEAM Plus and LEED. Acoustic commission tests for Reverberation Time shall be undertaken by a certified acoustic testing laboratory with The Hong Kong Laboratory Accreditation Scheme (HOKLAS) employed by the Contractor.

III. OUTLINE SPECIFICATION

A. Perforated Acoustic Metal Ceiling System

Specification of Perforated Acoustic Metal Ceiling System	
Material	Perforated acoustic metal ceiling with minimum 25mm thick acoustic insulation backing to provide minimum NRC 0.85, similar to “SAS Mesh”, “HunterDouglas Strength Mesh”, “Rockfan Metalscapes Wire Mesh” or equivalent.
Finish/color	As specified by the architect.
Inflammability	Comply with local fire regulation requirement, or as specified by the architect.
Acoustic Panel Extent	1. Refers to the architectural drawing. 2. Room with office type premises (at least 70% of total ceiling surface area): Security Control Room & Management Offices / Network Operation Centre / Office
Acoustical Performance	The acoustical performance shall meet or exceed: <u>Octave</u> 125 250 500 1000 2000 4000 Sound Absorption Coefficient ASTM C423 0.40 0.70 0.85 0.80 0.90 0.90

B. Acoustic Light Ceiling System

Specification of Acoustic Light Ceiling System	
Material	Acoustic light ceiling with minimum 20mm thick acoustic insulation backing to provide minimum NRC 0.70, similar to “Barrisol Acoustic Light”, or equivalent.
Finish/color	As specified by the architect.
Inflammability	Comply with local fire regulation requirement, or as specified by the architect.
Acoustic Panel Extent	1. Refers to the architectural drawing. 2. Conference Rooms: at least cover area directly above table and chairs
Acoustical Performance	The acoustical performance shall meet or exceed: <u>Octave</u> <u>125</u> <u>250</u> <u>500</u> <u>1000</u> <u>2000</u> <u>4000</u> Sound Absorption Coefficient ASTM C423 0.15 0.55 0.85 0.60 0.85 0.60

C. Acoustic Plasterboard Ceiling System

Specification of Acoustic Plasterboard Ceiling System	
Material	Perforated plasterboard ceiling with minimum 25mm thick acoustic insulation backing to provide minimum NRC 0.55, similar to “VoglToptec” or equivalent.
Finish/color	As specified by the architect.
Inflammability	Comply with local fire regulation requirement, or as specified by the architect.
Acoustic Panel Extent	1. Refers to the architectural drawing. 2. Conference Rooms: fully on the ceiling around the acoustic light.
Acoustical Performance	The acoustical performance shall meet or exceed: <u>Octave</u> <u>125</u> <u>250</u> <u>500</u> <u>1000</u> <u>2000</u> <u>4000</u> Sound Absorption Coefficient ASTM C423 0.36 0.47 0.55 0.55 0.54 0.54

III. SUBMITTAL REQUIREMENTS

The following is the minimum required for review prior to acceptance of Project Manager:

- A. Drawing showing size and fixing detail of acoustic panel.
- B. Acoustic performance laboratory test data.
- C. Product cutsheet and job reference.

IV. EXECUTION

- A. Bring to the Project Manager's attention prior to installation any conflicts with other trades in order to avoid inadequate space or other unforeseen conditions. Corrective work necessitated by conflicts after installation shall be at the responsible Contractor's expense.
- B. Obtain inspection and acceptance of any installation to be covered or enclosed.
- C. Correct, at no additional cost, all installations which are deemed defective in workmanship or materials.
- D. Fixing details and construction shall follow the manufacturer's specification and guideline and shall be approved by the Architect and the project acoustic consultant.

END OF SECTION